

LECTURE 6: INTRODUCTION TO HIERARCHICAL MODELS

Key concepts:

- Hyperparameters
- Exchangeable random variables
- de Finetti's Theorem
- Pooling
- Shrinkage

1. INTRODUCTION

One of the advantages of modern Bayesian methods is the ability to handle models with large numbers of parameters without “over-fitting.” In Bayesian hierarchical models, we model the data using as many parameters as we wish and then give these parameters a distribution, which depends upon a number of *hyperparameters*. This results in a “multilevel” model. At the lowest level, the distribution of the data depend upon a set of parameters. At the next level, the first level parameters depend upon a set of hyperparameters. We can continue in this fashion adding more levels of hyperparameters as necessary (though more than three levels of parameters is rare).

Some problems that lend themselves to this treatment include:

Cluster sampling. A common sampling design is to select clusters of units (often geographic areas) and then to subsample units within the selected first stage units. For example, one might first draw a collection of geographic areas. At the second stage, people are selected from the geographic areas selected in the first stage. Because of differences between geographic areas, it would be desirable to have separate parameters for each cluster, though of course these parameters tend to be similar across clusters.

Panel data. In panel surveys, the same respondents are interviewed multiple times. The panel units can be persons, firms, countries, *etc.* Because respondents are heterogeneous, we would like to allow each panel unit to have its own parameters.

Metaanalysis. In a metaanalysis, the same question is addressed using multiple different datasets (usually by different researchers). Each dataset has been used previously to estimate parameters of a model. Because of differences between the methods of data collection, models, *etc.*, the estimates will vary. In synthesizing the data, we wish to estimate the hyperparameters. This allows us to describe common findings as well as variation between the various studies.

We will consider many other use of hierarchical models, but this is enough to get us started.

2. EXCHANGEABILITY

One justification for hierarchical modeling comes from de Finetti's Theorem.

First, some definitions.

Definition 1. A set of random variables $X_1, \dots, X_n$ are said to be exchangeable if $X_{\pi(1)}, \dots, X_{\pi(n)}$ has the same distribution as $X_1, \dots, X_n$ for any permutation π mapping $\{1, \dots, n\}$ to itself.

Exchangeability is a symmetry condition, meaning that we can't distinguish between the observations. An i.i.d. sequence is exchangeable, though the converse is not true. Exchangeable random variables are identically distributed, but may be dependent. (In fact, (X, X) is obviously exchangeable.) We need a somewhat stronger version of exchangeability for de Finetti's Theorem.

Definition 2. A sequence of random variables $X_1, X_2, \dots$ is infinitely exchangeable if $X_1, \dots, X_n$ are exchangeable for all n .

Theorem 1 (de Finetti). Suppose $X_1, X_2, \dots$ is an infinitely exchangeable set of dichotomous variables. Then there exists a distribution function F with support on $[0, 1]$ such that

$$\mathbb{P}\{X_1 = x_1, \dots, X_n = x_n\} = \int_0^1 \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i} dF(\theta)$$

for $x_i \in \{0, 1\}$ for $i = 1, \dots, n$, and $n = 1, 2, \dots$.

Proof. See Heath and Sudderth, “de Finetti's Theorem on Exchangeable Variables,” *American Statistician* (1976). $\square$

de Finetti's Theorem justifies modeling exchangeable dichotomous data as i.i.d. conditional upon a parameter θ with prior distribution F .

The Hewitt-Savage Theorem generalizes de Finetti's Theorem. (I'm omitting some technical conditions in Hewitt and Savage, “Symmetric Measures on Cartesian Products,” *Transactions of the American Mathematical Society*, 1955.)

Theorem 2 (Hewitt-Savage). Let $X_1, X_2, \dots$ be an infinitely exchangeable sequence of random variables. Then there exists a measure μ defined over probability measures Q such that

$$\mathbb{P}\{X_1 \in A_1, \dots, X_n \in A_n\} = \int \prod_{i=1}^n Q(A_i) \mu(dQ)$$

for events $A_1, \dots, A_n$ and $n = 1, 2, \dots$.

Again, this provides a justification for modeling exchangeable random variables as i.i.d. conditional upon an unknown “parameter” Q with prior distribution μ .

Diaconis and Freedman, “Finite Exchangeable Sequences,” *Annals of Probability* (1980), prove some approximate results without assuming infinite exchangeability.

3. BAYESIAN ANALYSIS OF A HIERARCHICAL BINOMIAL MODEL

Suppose $y_j \sim \text{Binomial}(n_j, \theta_j)$ and $\theta_j \sim \text{Beta}(\alpha, \beta)$ for $j = 1, \dots, J$ so

$$p(\theta|\alpha, \beta) \propto \prod_{j=1}^J p(\theta_j|\alpha, \beta) \propto p(\alpha, \beta) \prod_{j=1}^J \frac{\theta_j^{\alpha-1} (1 - \theta_j)^{\beta-1}}{B(\alpha, \beta)}$$

For the moment, we leave the (marginal) prior $p(\alpha, \beta)$ for the hyperparameters unspecified. The likelihood is

$$p(y_1, \dots, y_J | \alpha, \beta, \theta) = \prod_{j=1}^J \binom{n_j}{y_j} \theta^{y_j} (1 - \theta)^{n_j - y_j}$$

The joint posterior is

$$\begin{aligned} p(\alpha, \beta, \theta | y) &\propto p(\alpha, \beta) p(\theta | \alpha, \beta) p(y | \theta) \\ &\propto p(\alpha, \beta) \prod_{j=1}^J \frac{\theta_j^{\alpha-1} (1 - \theta_j)^{\beta-1}}{B(\alpha, \beta)} \prod_{j=1}^J \theta_j^{y_j} (1 - \theta_j)^{n_j - y_j} \end{aligned}$$

The conditional posterior for θ is

$$\begin{aligned} p(\theta | \alpha, \beta, y) &= \frac{p(\alpha, \beta, \theta | y)}{p(\alpha, \beta | y)} \\ &\propto \frac{p(\alpha, \beta)}{p(\alpha, \beta | y) B(\alpha, \beta)^n} \prod_{j=1}^J \theta_j^{\alpha+y_j-1} (1 - \theta_j)^{\beta+n_j-y_j}. \end{aligned}$$

As a function of $\theta_1, \dots, \theta_J$ (you need to be careful about this!),

$$p(\theta_1, \dots, \theta_J | \alpha, \beta, y) \propto \prod_{j=1}^J \theta_j^{\alpha+y_j-1} (1 - \theta_j)^{\beta+n_j-y_j},$$

showing that, conditional upon α, β , and y , we have $\theta_j \sim \text{Beta}(\alpha + y_j, \beta + n_j - y_j)$ as independent conditional posteriors. That is,

$$p(\theta | \alpha, \beta, y) = \prod_{j=1}^J \frac{1}{B(\alpha + y_j, \beta + n_j - y_j)} \theta_j^{\alpha+y_j-1} (1 - \theta_j)^{\beta+n_j-y_j}.$$

This roundabout argument is necessary, since the next result depends on having the posterior pdf of θ , including its normalizing constant (because it depends on the hyperparameters α and β).

To find the marginal posterior for the hyperparameters, use

$$p(\alpha, \beta, \theta | y) = p(\alpha, \beta | y) p(\theta | \alpha, \beta, y)$$

so, ignoring terms that don't involve α and β ,

$$\begin{aligned} p(\alpha, \beta | y) &= \frac{p(\alpha, \beta, \theta | y)}{p(\theta | \alpha, \beta, y)} \\ &\propto \frac{\frac{p(\alpha, \beta)}{B(\alpha, \beta)^n} \prod_{j=1}^J \theta_j^{\alpha+y_j-1} (1 - \theta_j)^{\beta+n_j-y_j}}{\prod_{j=1}^J \frac{1}{B(\alpha + y_j, \beta + n_j - y_j)} \theta_j^{\alpha+y_j-1} (1 - \theta_j)^{\beta+n_j-y_j}} \\ &\propto p(\alpha, \beta) \prod_{j=1}^J \frac{B(\alpha + y_j, \beta + n_j - y_j)}{B(\alpha, \beta)}. \end{aligned}$$

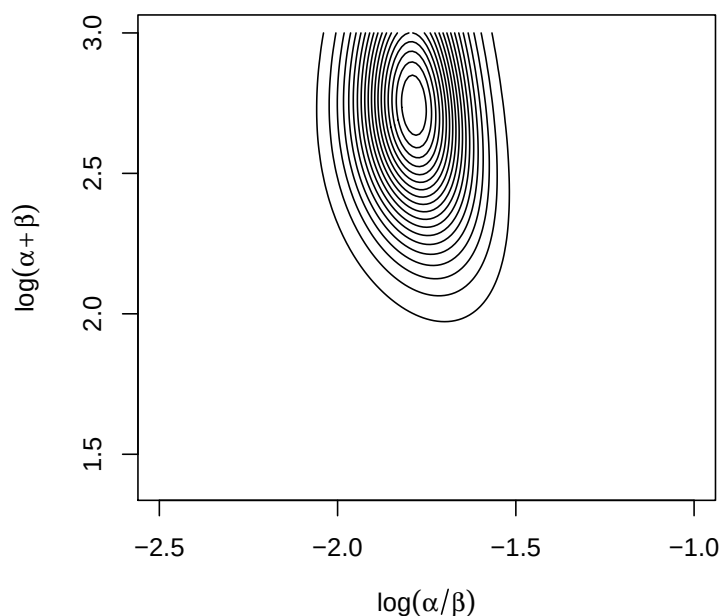


FIGURE 1. Contours of the marginal posterior distribution of the hyperparameters. The contours shown are for levels ranging from 5% to 95% of the maximum of the posterior density.

Several possible choices for the prior for the hyperparameters are discussed in *BDA3*, some of which work (*i.e.*, lead to a proper posterior) and some of which don't. Gelman *et al.* (2013), p. 110 suggest using

$$p(\alpha, \beta) \propto (\alpha + \beta)^{-5/2}$$

We will discuss other choices later, but this one doesn't seem to contain much information.

For the rat tumor data, here is the posterior density for the hyperparameters. The parameters have been transformed from $(\alpha, \beta) \mapsto (\log(\alpha/\beta), \log(\alpha + \beta))$ (referred to in the R code as `logodds` and `logprec`).

Here is the corresponding R script.

```
df <- read.table("http://www.stat.columbia.edu/~gelman/book/data/rats.asc",
  skip=3, header=TRUE)
y <- df$y; n <- df$N; J <- nrow(df)

# Figures 5.2 in BDA3 (using transformed parameters)
logq <- function(a,b) log(a*b) - 2.5 * log(a+b) + sum(lbeta(a+y,b+n-y) - lbeta(a,b))
logodds <- seq(-2.5, -1.0, .01); logprec <- seq(1.4, 3.0, .01)
posterior <- matrix(nrow=length(logodds), ncol=length(logprec))
for (i in 1:length(logodds)) {
  for (j in 1:length(logprec)) {
```

```

        a <- exp(logodds[i]+logprec[j]) / (1 + exp(logodds[i]))
        b <- exp(logprec[j]) / (1 + exp(logodds[i]))
        posterior[i,j] <- logq(a,b)
    }
}
pdf(file="lec06-fig01.pdf", width=5, height=5)
contour(logodds, logprec, posterior, levels=seq(.05,.95,.05), drawlabels=FALSE,
        xlab=expression(log(alpha/beta)), ylab=expression(log(alpha+beta)))

```

4. COMPUTATION

Soon we will have MCMC methods that will make short work of this posterior distribution, but for now consider the following simple strategy:

1. Compute the kernel of $p(\alpha, \beta|y)$ on a grid of points $G = \{(\alpha_k, \beta_\ell) : k = 1, \dots, L; \ell = 1, \dots, L\}$. Call this unnormalized marginal posterior q ,

$$q(\alpha, \beta) = (\alpha + \beta)^{-5/2} \prod_{j=1}^J \frac{B(\alpha + y_j, \beta + n_j - y_j)}{B(\alpha, \beta)}.$$

2. Normalize $q(\alpha_k, \beta_\ell)$ on G by dividing by the sum,

$$\tilde{q}(\alpha_k, \beta_\ell) = \frac{q(\alpha_k, \beta_\ell)}{\sum_{k=1}^K \sum_{\ell=1}^L q(\alpha_k, \beta_\ell)}$$

This gives a discrete probability distribution over G .

3. Draw M values of the hyperparameters from $\tilde{q}$: $(\alpha^{(1)}, \beta^{(1)}), \dots, (\alpha^{(M)}, \beta^{(M)})$.
4. For each $(\alpha^{(m)}, \beta^{(m)})$ generated in step 3, simulate a value $\theta^{(m)} = (\theta_1^{(m)}, \dots, \theta_n^{(m)})$ using $\theta_j | \alpha = \alpha^{(m)}, \beta = \beta^{(m)}, y \sim \text{Beta}(\alpha^{(m)} + y_j, \beta^{(m)} + n_j - y_j)$.

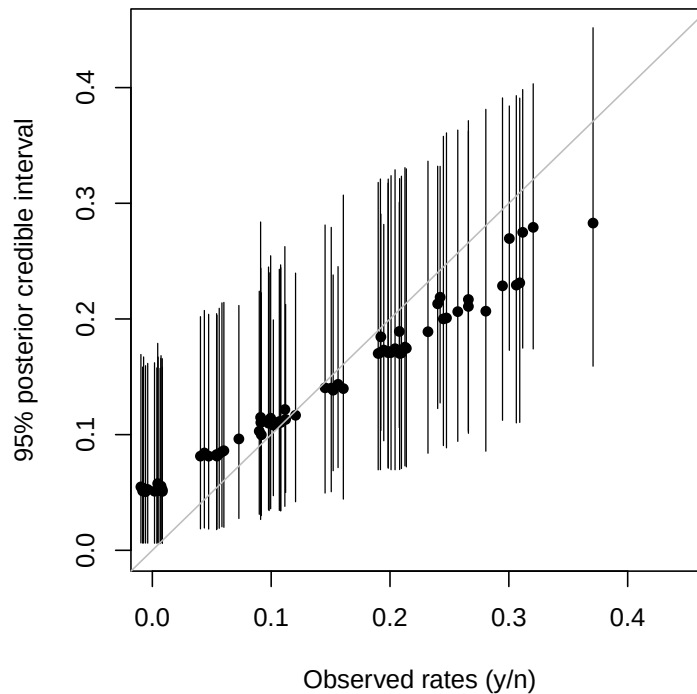


FIGURE 2. Comparison of the sample proportions with the posterior mean of θ_j in each experiment. Note the shrinkage, so that the posterior mean probability in experiments with high or low sample proportions is closer to the overall average.

Here is R code to find the posterior means for the θ_j 's and plot them against the sample proportions.

```
# marginal unnormalized posterior for hyperparameters
logq <- function(a,b) -2.5 * log(a+b) + sum(lbeta(a+y,b+n-y) - lbeta(a,b))

# 1: make a grid G and evaluate q at the grid points
G <- expand.grid(a=seq(0.3,5.4,.1), b=seq(2.9,18.5,.1))
qtilde <- sapply(1:nrow(G), function(i) logq(G$a[i], G$b[i]))
qtilde <- exp(qtilde - max(qtilde))

# 2. normalize q
qtilde <- qtilde / sum(qtilde)

# 3. draw M pairs of hyperparameters
M <- 10000
phi <- G[sample(nrow(G), size=M, replace=TRUE, prob=qtilde),]

# 4. generate M values of (theta1,...,thetaJ)
```

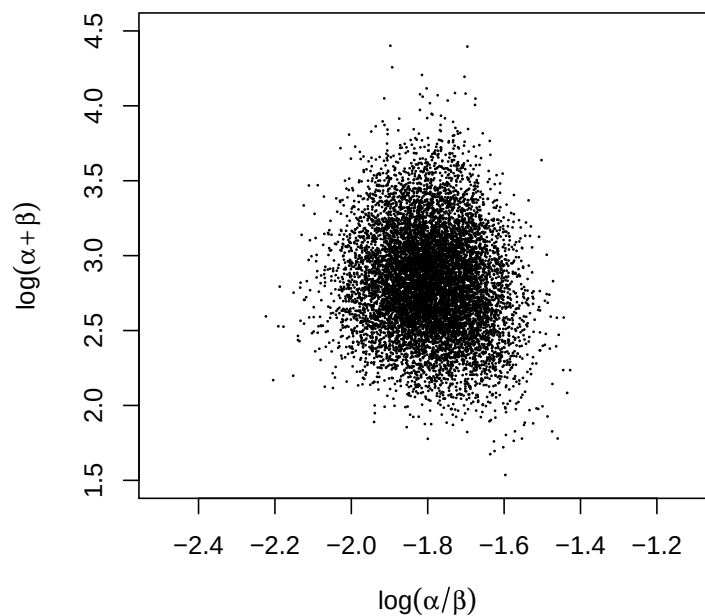


FIGURE 3. 10,000 draws from the posterior distribution of $\log(\alpha/\beta)$ and $\log(\alpha+\beta)$ using JAGS.

```
theta <- as.matrix(do.call(rbind, lapply(1:M, function(i)
  rbeta(J, shape1=phi$a[i]+y, shape2=phi$b[i]+n-y))))

# Figure 5.4 from BDA3
medians <- apply(theta, 2, median)
lo <- apply(theta, 2, quantile, probs=.025)
hi <- apply(theta, 2, quantile, probs=.975)
means <- jitter(y/n, amount=.01)
plot(x=means, y=medians, pch=19, cex=0.8, xlim=c(0,.45), ylim=c(0,.45),
     xlab="Observed rates (y/n)", ylab="95% posterior credible interval",
     main="")
for (i in 1:length(means)) lines(x=rep(means[i],2), y=c(lo[i], hi[i]), lwd=.8)
abline(0, 1, col="grey")
```

5. MCMC

As a preview of things to come, here is how you could perform the same analyses in JAGS, *sans* algebra.

```
library(rjags)
model <- "model {
  for (j in 1:J) {
```

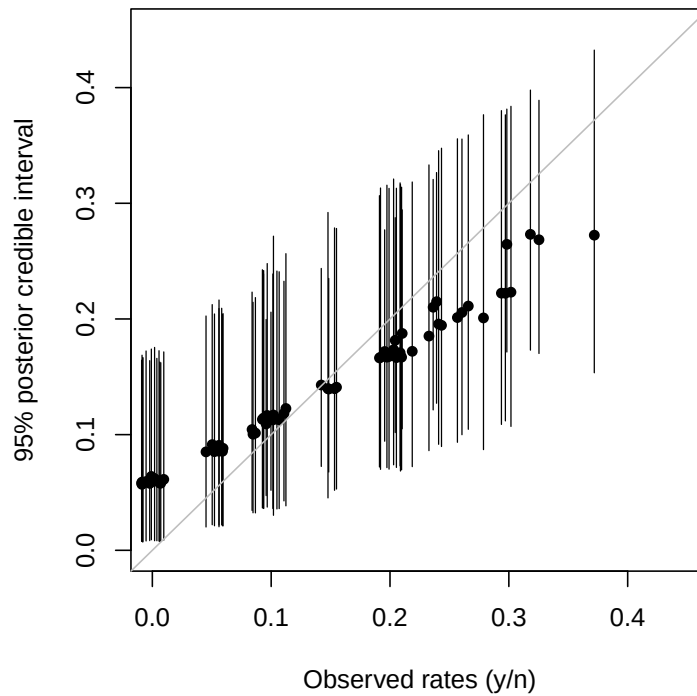


FIGURE 4. Comparison of the sample proportions with the posterior mean of θ_j in each experiment estimated using JAGS.

```

    y[j] ~ dbinom(theta[j], n[j])
    theta[j] ~ dbeta(alpha, beta)
  }
  alpha <- exp(logodds+logprec) / (1 + exp(logodds))
  beta <- exp(logprec) / (1 + exp(logodds))
  logodds ~ dunif(-3, 0)
  logprec ~ dunif(0, 10)
}"
data <- list(y=y, n=n, J=J)
sampler <- jags.model(textConnection(model), data=data)
update(sampler, 1000)
samples <- jags.samples(sampler, variable.names=c("alpha", "beta", "theta"),
  n.iter=10000)
logodds <- with(samples, log(alpha/beta))
logprec <- with(samples, log(alpha+beta))
plot(x=logodds, y=logprec, pch=19, cex=0.1, xlim=c(-2.5, -1.1),
  ylim=c(1.5, 4.5), xlab=expression(log(alpha/beta)),
  ylab=expression(log(alpha+beta)), main="")
means <- jitter(y/n, amount=.01)
medians <- summary(samples$theta, median)$stat

```



```

lo <- summary(samples$theta, quantile, prob=.025)$stat
hi <- summary(samples$theta, quantile, prob=.975)$stat
plot(x=means, y=medians, pch=19, cex=0.8, xlim=c(0,.45), ylim=c(0,.45),
     xlab="Observed rates (y/n)", ylab="95% posterior credible interval",
     main="")
for (i in 1:length(means)) lines(x=rep(means[i],2), y=c(lo[i], hi[i]), lwd=.8)
abline(0, 1, col="grey")

```